



RV College of
Engineering®

VII Semester Computer Graphics & Virtual Reality (CS373IA)

Department of Computer Science & Engineering

RV College of Engineering, Bengaluru.

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Unit- 4

Viewing and Visible Surface Detection

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Unit –IV	9 Hrs
Geometric Transformations (contd..) 3-D Geometrical Transformations: Translation, Scaling, Rotation. Viewing and Visible Surface Detection: Viewing pipeline: viewing coordinates, Aspect Ratio and view ports, view volume, 3-D clipping. Projections: Classification of planar geometric projections, Projections in OpenGL; Visible surface detection: Classification, back-face detection, depth-buffer, scan-line, depth sorting, BSP-tree methods, area sub-division and octree methods.	



Viewing and Visible Surface Detection: Viewing pipeline: viewing coordinates, Aspect Ratio and view ports, view volume, 3-D clipping.

Projections: Classification of planar geometric projections, Projections in openGL;

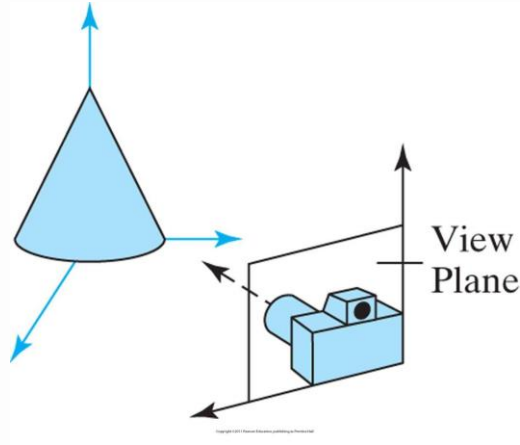
Visible surface detection: Classification, back-face detection, depth-buffer, scan-line, depth sorting, BSP-tree methods, area sub-division and octree methods.



3D Viewing:

- Procedures for displaying views of a three-dimensional scene on an output device involve many aspects:
 - Generate a 3D scene with objects generally defined with a set of surfaces forming a closed boundary around the object interior.
 - Generate interior of 3D objects, if needed.
 - Project object surfaces views onto the surface of a display device with 3D viewing pipeline.
 - Identification of visible parts.
 - Lighting effects and surface characteristics.

3D Viewing (Contd.):



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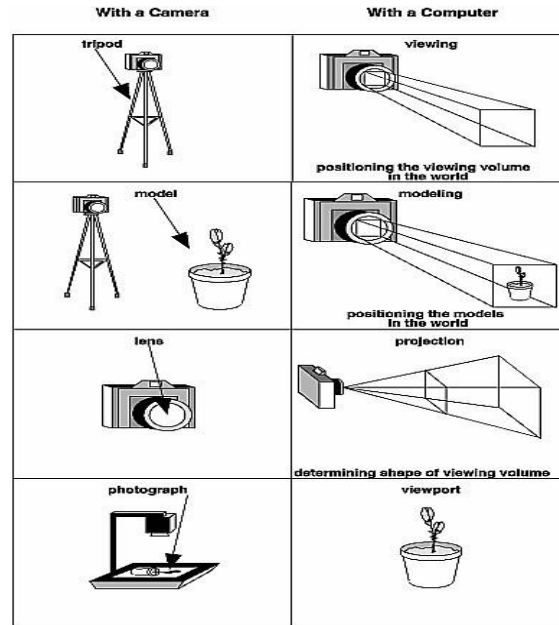
Coordinate reference for obtaining a selected view of a three-dimensional scene.



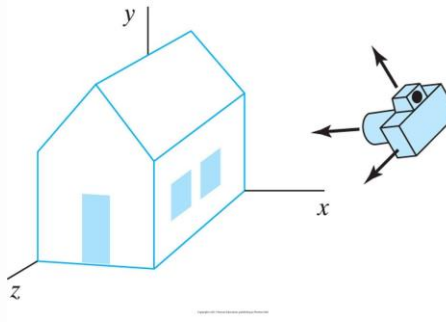
3D Viewing - Camera Analogy :

- The transformation process to produce the desired scene for viewing is analogous to taking a photograph with a camera. The steps with a camera (or a computer) might be the following.
 1. Set up your tripod and pointing the camera at the scene (viewing transformation)
 2. Arrange the scene to be photographed into the desired composition (modeling transformation)
 3. Choose a camera lens or adjust the zoom (projection transformation)
 4. Determine how large you want the final photograph to be - for example, you might want it enlarged (viewport transformation)⁷
- After these steps are performed, the picture can be snapped or the scene can be drawn.

3D Viewing - Camera Analogy : (Contd..)



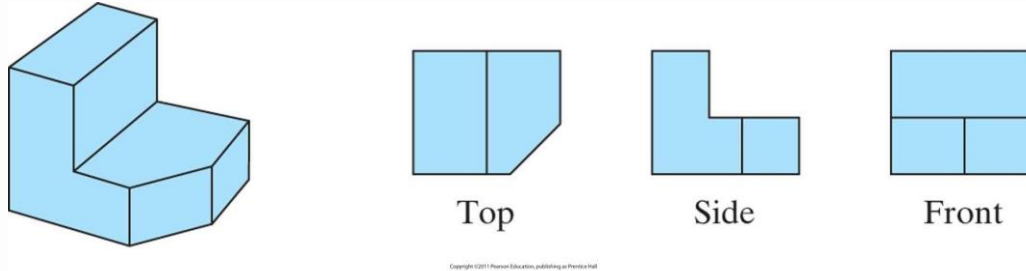
3D Viewing - Photographing



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Photographing a scene involves selection of the camera position and orientation

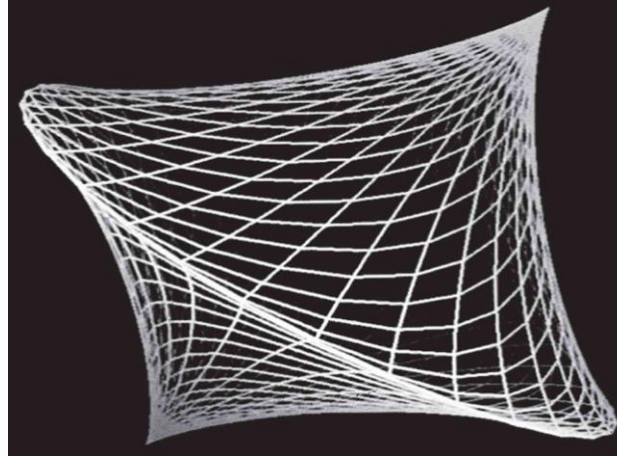
Projections



Three parallel-projection views of an object, showing relative proportions from different viewing positions.



Wire-frame Representation & Depth Cueing



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A wire-frame object displayed with depth cueing, so that the brightness of lines decreases from the front of the object to the back.

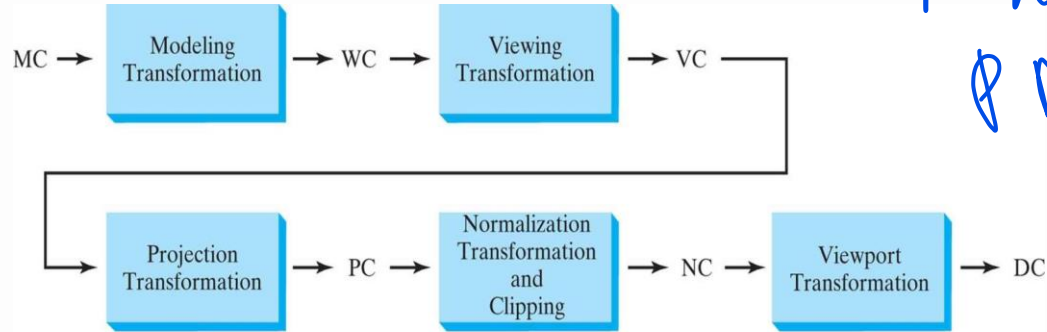


Other 3D Viewing Concepts

- Visible line and surface detection
- Surface rendering
- Exploded and cutaway views
- 3D and stereoscopic viewing



3D Viewing Pipelining

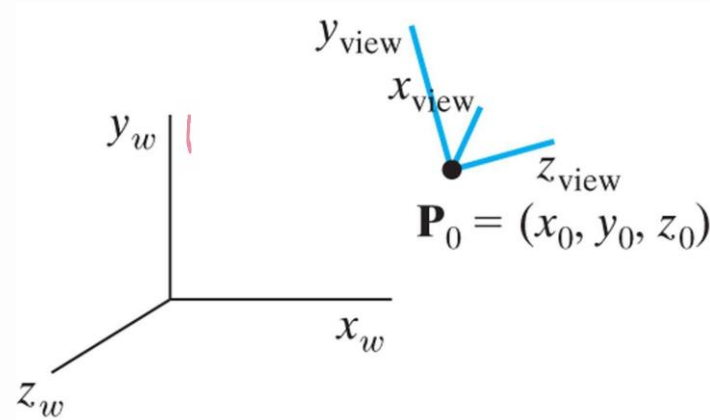


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& ND

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General three-dimensional transformation pipeline, from modeling coordinates (MC) to world coordinates (WC) to viewing coordinates (VC) to projection coordinates (PC) to normalized coordinates (NC) and, ultimately, to device coordinates (DC).

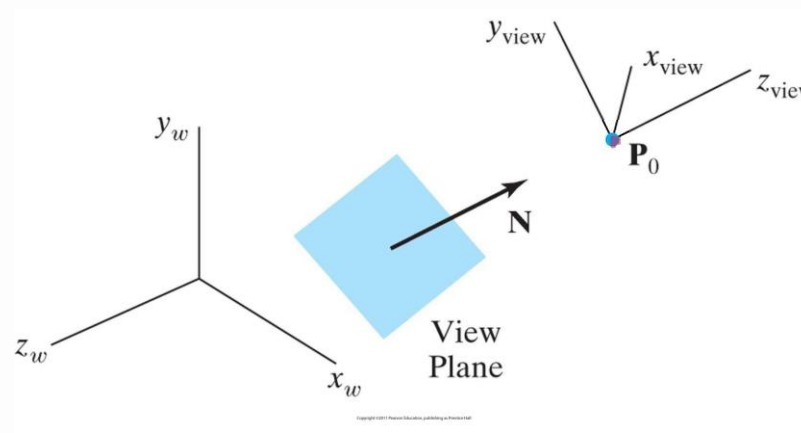
3D Viewing Coordinate Parameters: The View-Plane Normal Vector



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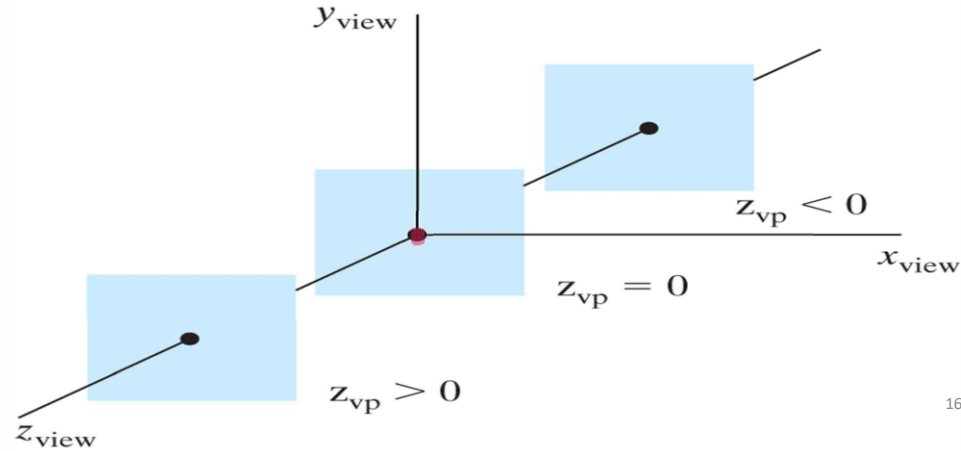
A right-handed viewing-coordinate system, with axes x_{view} , y_{view} , and z_{view} , relative to a right-handed world-coordinate frame.

3D Viewing Coordinate Parameters: The View-Plane Normal Vector (Cont..)



Orientation of the view plane and view-plane normal vector $\mathbf{N}$

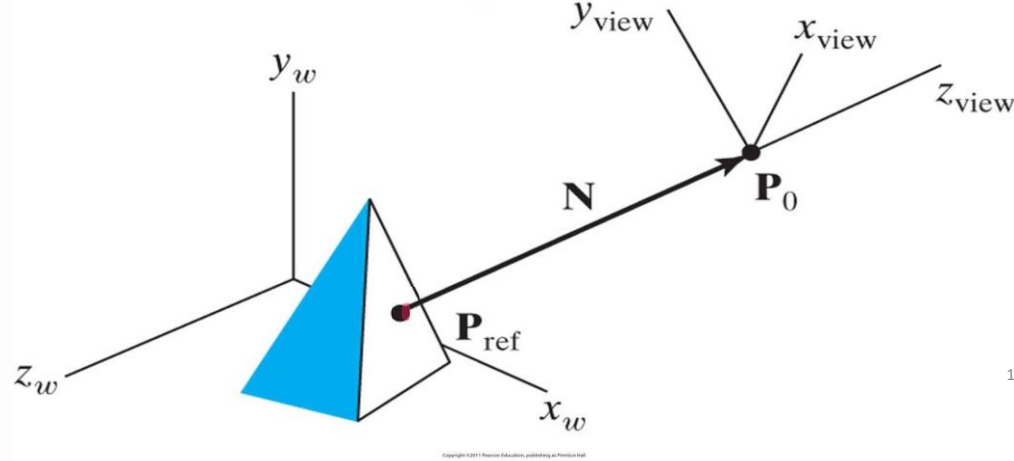
3D Viewing Coordinate Parameters: The View-Plane Normal Vector (Cont..)



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Three possible positions for the view plane along the z_{view} axis.

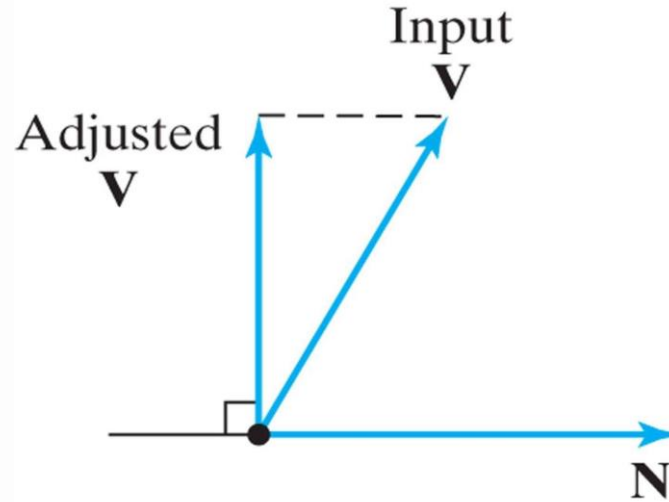
3D Viewing Coordinate Parameters: The View-Plane Normal Vector (Cont..)



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Specifying the view-plane normal vector N as the direction from a selected reference point P_{ref} to the viewing-coordinate origin P_0 .

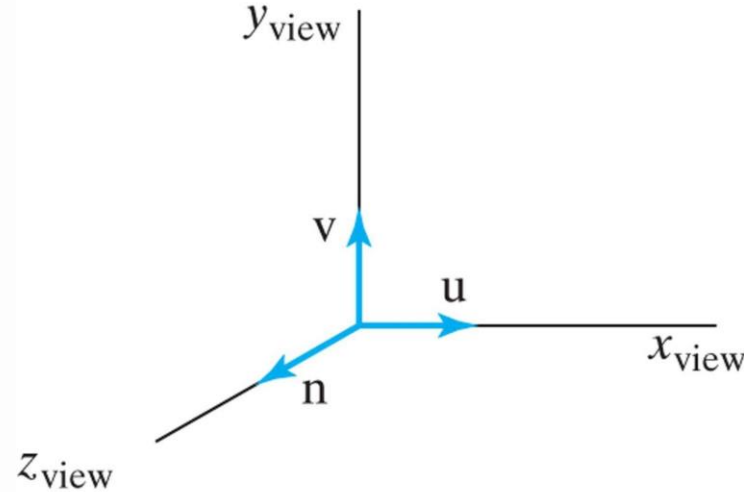
3D Viewing Coordinate Parameters: The View-Up Vector



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Adjusting the input direction of the view-up vector $\mathbf{V}$ to an orientation perpendicular to the view-plane normal vector $\mathbf{N}$.

3D Viewing Coordinate Parameters: uvn Reference Frame



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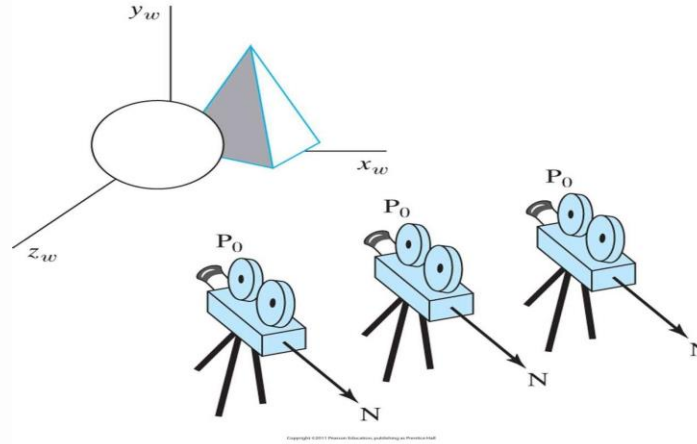
A right-handed viewing system defined with unit vectors u , v , and n .



Modeling Transformation

- Use the modeling transformation to position and orient the model
 - Rotate, scale and translate the model or perform some combination of these operations
 - `glScalef(1.0, 2.0, 1.0);`
- Note that instead of moving the camera (with a **viewing transformation**) so that the cube could be viewed, you could have moved the cube away from the camera (with a **modeling transformation**)

Transformation from World to Viewing Coordinates

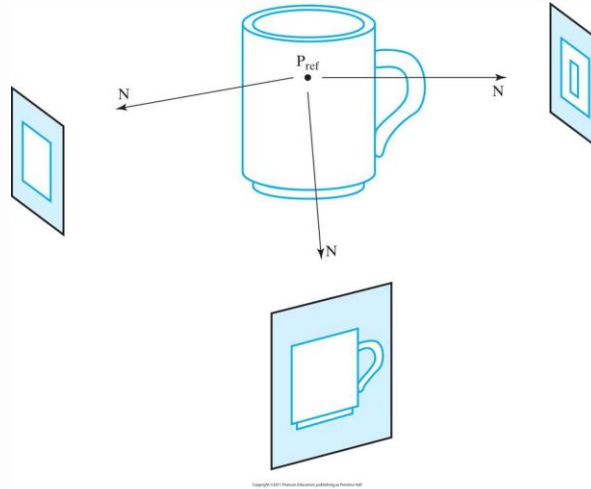


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Panning across a scene by changing the viewing position, with a fixed direction for N .



Transformation from World to Viewing Coordinates (Contd..)



22

Viewing an object from different directions using a fixed reference point.



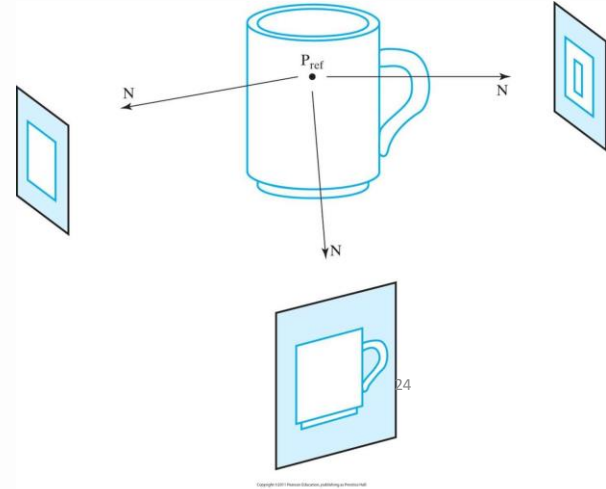
World to Viewing Coordinates (Contd..)

- How is it achieved?

Transformation from World to Viewing Coordinates (Contd..)

- **Accomplished using** the methods for transforming between coordinate system :
 1. Translate the viewing-coordinate origin to the origin of the world coordinate system.
 2. Apply rotations to align the x_{view} , y_{view} , and z_{view} axes with the world x_w, y_w , and z_w axes, respectively.
- The viewing-coordinate origin is at world position $P_0 = (x_0, y_0, z_0)$. Therefore, the matrix for translating the viewing origin to the world origin is

$$T = \begin{bmatrix} 1 & 0 & 0 & -x_0 \\ 0 & 1 & 0 & -y_0 \\ 0 & 0 & 1 & -z_0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Transformation from World to Viewing Coordinates (Contd..)

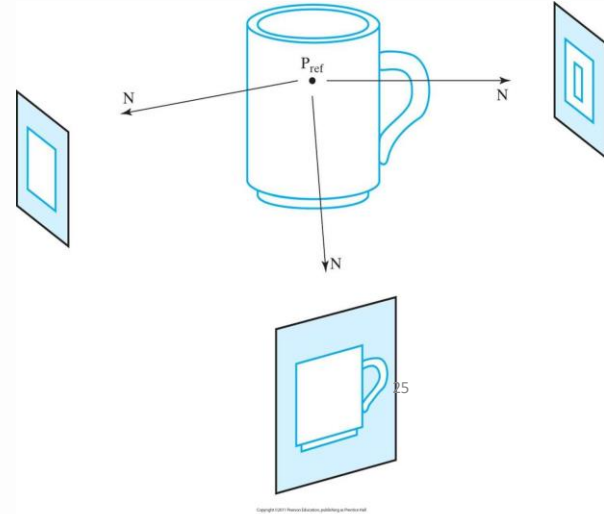
- For the rotation transformation, use the unit vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{n}$ to form the composite rotation matrix that superimposes the viewing axes onto the world frame. This transformation matrix is

$$\mathbf{R} = \begin{bmatrix} u_x & u_y & u_z & 0 \\ v_x & v_y & v_z & 0 \\ n_x & n_y & n_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

where the elements of matrix $\mathbf{R}$ are the components of the $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{n}$ axis vectors.

- The coordinate transformation matrix is then obtained as the product of the preceding translation and rotation matrices:

$$\begin{aligned} \mathbf{M}_{WC, VC} &= \mathbf{R} \cdot \mathbf{T} \\ &= \begin{bmatrix} u_x & u_y & u_z & -\mathbf{u} \cdot \mathbf{P}_0 \\ v_x & v_y & v_z & -\mathbf{v} \cdot \mathbf{P}_0 \\ n_x & n_y & n_z & -\mathbf{n} \cdot \mathbf{P}_0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$



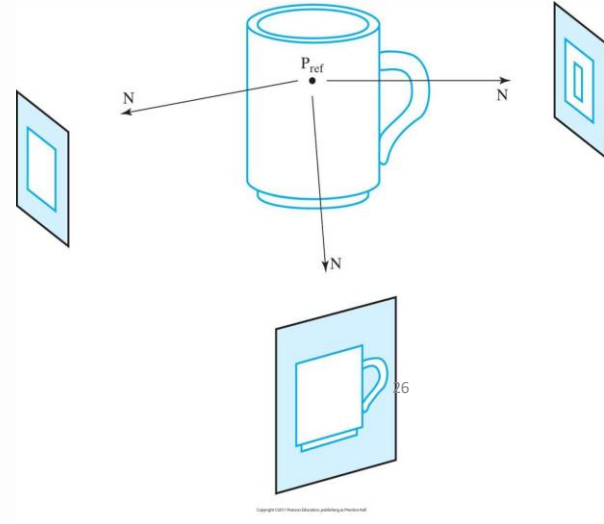
Transformation from World to Viewing Coordinates (Contd..)

- Translation factors in this matrix are calculated
 - as the vector dot product of each of the $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{n}$ unit vectors with $\mathbf{P}_0$, which represents a vector from the world origin to the viewing origin.
 - In other words, the translation factors are the negative projections of $\mathbf{P}_0$ on each of the viewing-coordinate axes (the negative components of $\mathbf{P}_0$ in viewing coordinates).
 - These matrix elements are evaluated as

$$-\mathbf{u} \cdot \mathbf{P}_0 = -x_0u_x - y_0u_y - z_0u_z$$

$$-\mathbf{v} \cdot \mathbf{P}_0 = -x_0v_x - y_0v_y - z_0v_z$$

$$-\mathbf{n} \cdot \mathbf{P}_0 = -x_0n_x - y_0n_y - z_0n_z$$





Projection Transformation

- Specifying the projection transformation is like choosing a lens for a camera
- You can think of this transformation as determining what the field of view or viewing volume is and therefore what objects are inside it
- This is equivalent to choosing among wide-angle, normal, and telephoto lenses, for example
 - With a wide-angle lens, you can include a wider scene in the final photograph than with a telephoto lens, but a telephoto lens allows you to²⁷ photograph objects as though they're closer to you than they actually are



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Geometric Projections

Projection

“An operation that transforms points in N-dimensional space to another of dimensionality *less than n*”

Geometric Projection

Application to geometric coordinate systems (Cartesian, Polar, Spherical and Cylindric systems)

Types of Projections

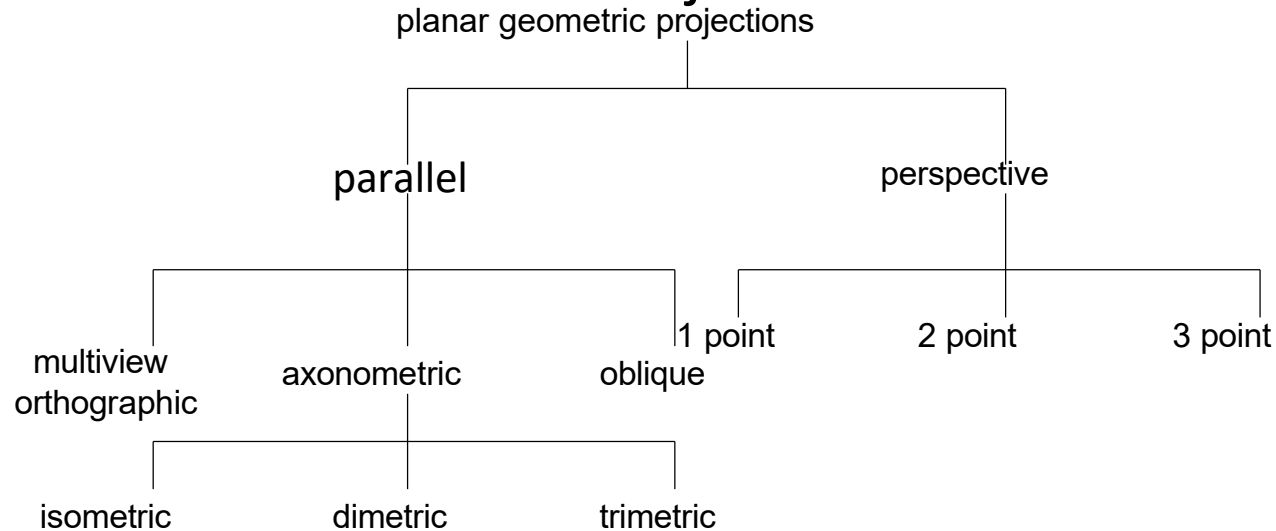
- Planar
- Non-Planar

Application to Computer Graphics

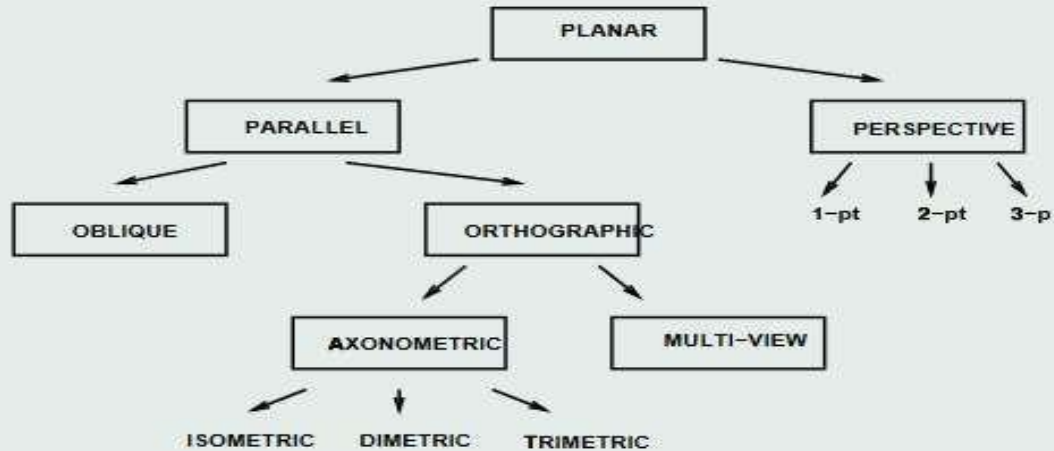
1. Viewing 3D environments on a 2D screen.
2. Visualization of multi-variate data from scientific applications.



Taxonomy of Planar Geometric Projections



Planar Projections



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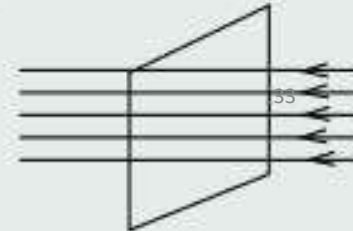
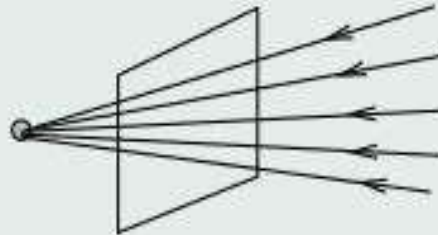


Parallel Projection

Projectors are parallel to each other.

Perspective

Projectors converge to a point.





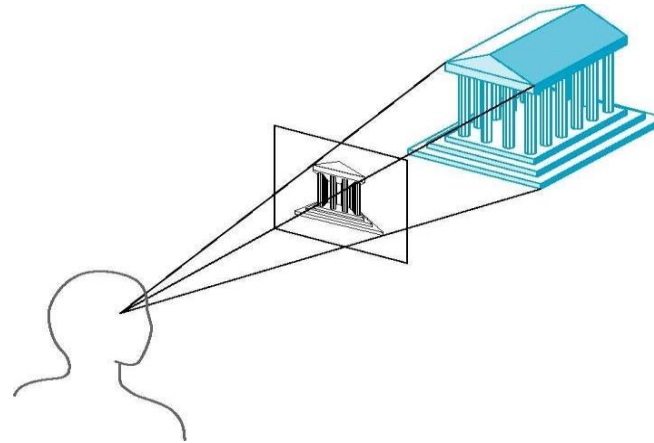
Planar Geometric Projections

- Standard projections project onto a plane
- Projectors are lines that either
 - converge at a center of projection
 - are parallel
- Such projections preserve lines
 - but not necessarily angles
- Nonplanar projections are needed for applications such as map cons

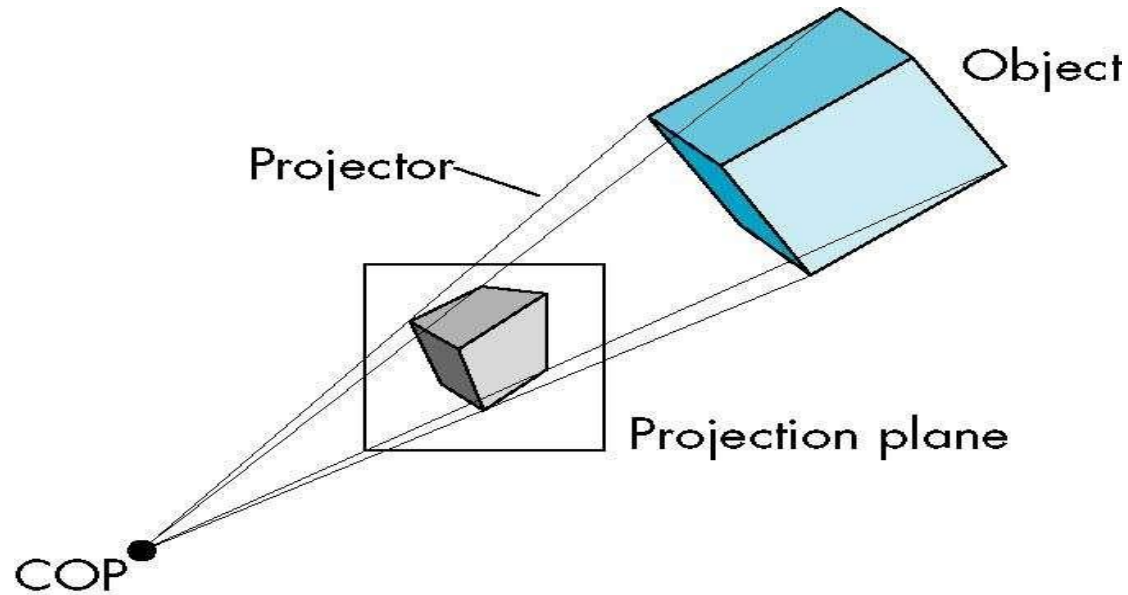


Perspective Projection

Projectors converge at center of projection

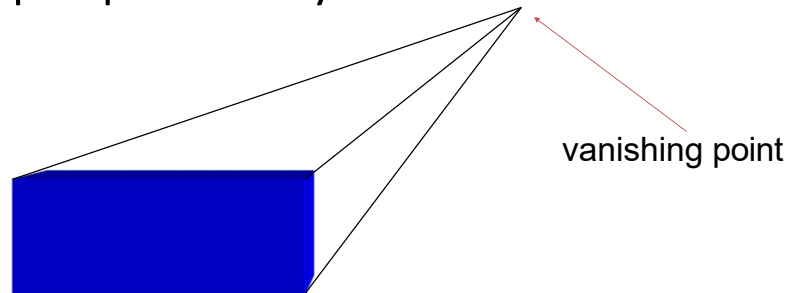


Perspective Projection



Vanishing Points

- Parallel lines (not parallel to the projection plan) on the object converge at a single point in the projection (the *vanishing point*)
- Drawing simple perspectives by hand uses these vanishing point(s)





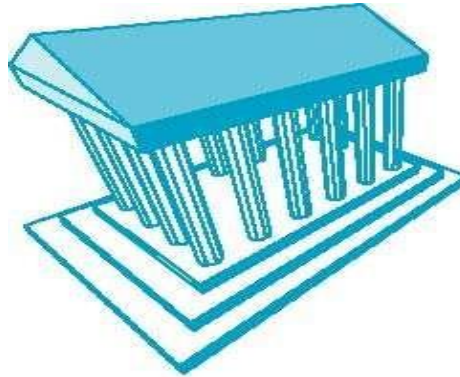
Perspective vs Parallel

- Computer graphics treats all projections the same and implements them with a single pipeline
- Classical viewing developed different techniques for drawing each type of projection
- Fundamental distinction is between parallel and perspective viewing even though mathematically parallel viewing is the limit of perspective viewing



Three-Point Perspective

- No principal face parallel to projection plane
- Three vanishing points for cube



Two-Point Perspective

- On principal direction parallel to projection plane
- Two vanishing points for cube



One-Point Perspective

- One principal face parallel to projection plane
- One vanishing point for cube

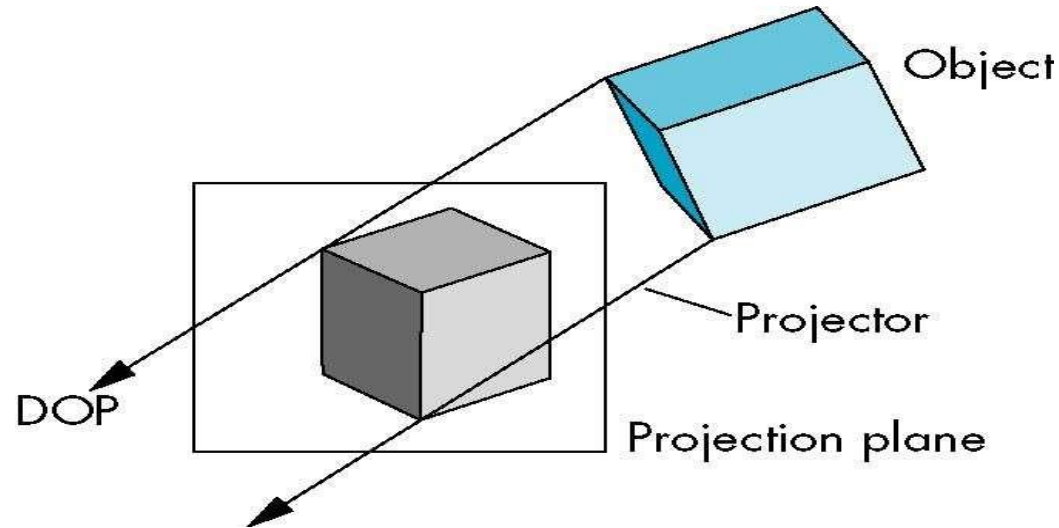




Advantages and Disadvantages

- Objects further from viewer are projected smaller than the same sized objects closer to the viewer (*diminution*)
 - Looks realistic
- Equal distances along a line are not projected into equal distances (*nonuniform foreshortening*)
- Angles preserved only in planes parallel to the projection plane
- More difficult to construct by hand than parallel projections
(but not more difficult by computer)

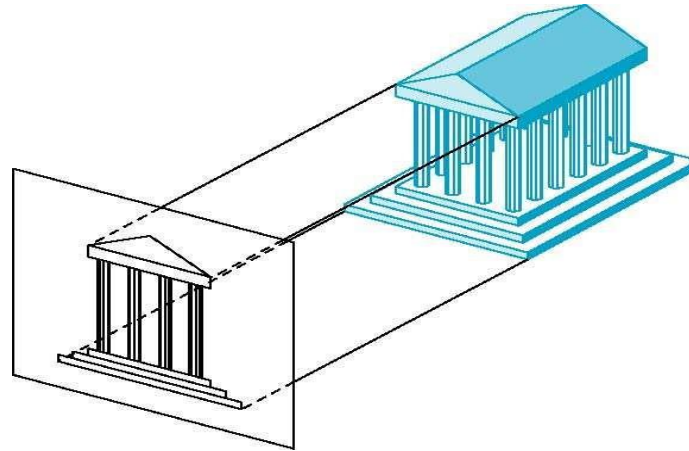
Parallel Projection





Orthographic Projection

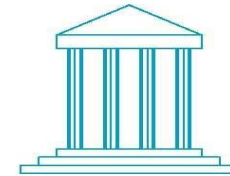
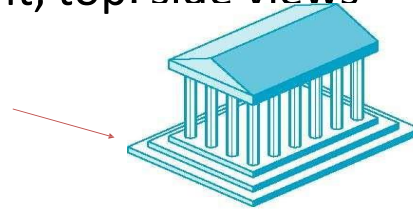
Projectors are orthogonal to projection surface



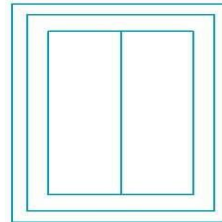
Multiview Orthographic Projection

- Projection plane parallel to principal face
- Usually form front, top, side views

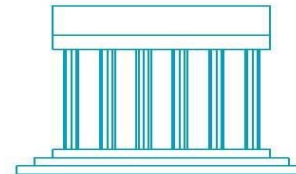
isometric (not multiview
orthographic view)



front



top



side

in CAD and architecture,
we often display three
multiviews plus isometric



Advantages and Disadvantages

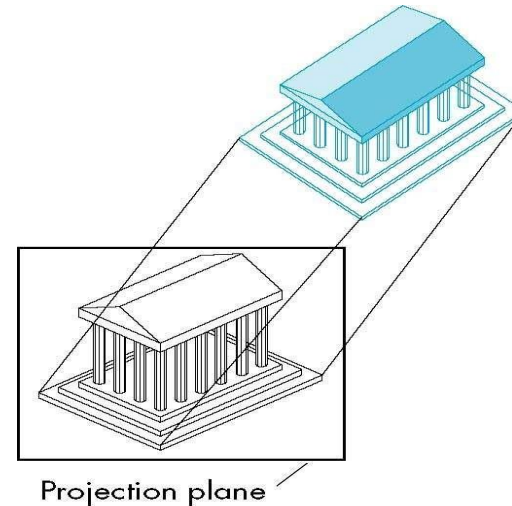
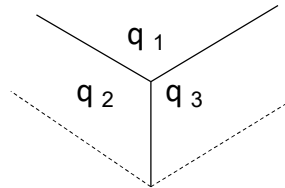
- Preserves both distances and angles
 - Shapes preserved
 - Can be used for measurements
 - Building plans
 - Manuals
- Cannot see what object really looks like because many surfaces hidden from view
 - Often we add the isometric

Axonometric Projections

Allow projection plane to move relative to object

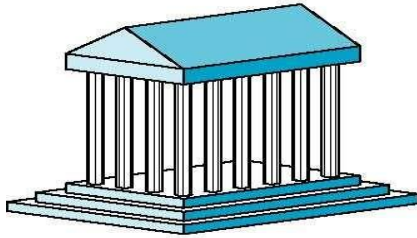
classify by how many angles of
a corner of a projected cube are
the same

none: trimetric
two: dimetric
three: isometric

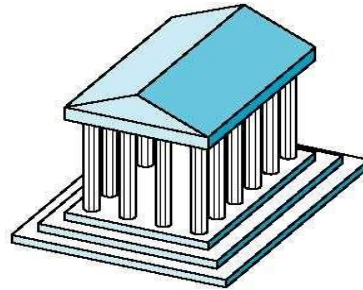




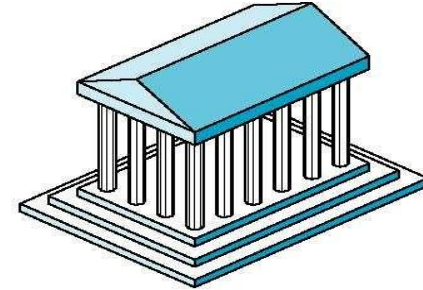
Types of Axonometric Projections



Dimetric



Trimetric



Isometric

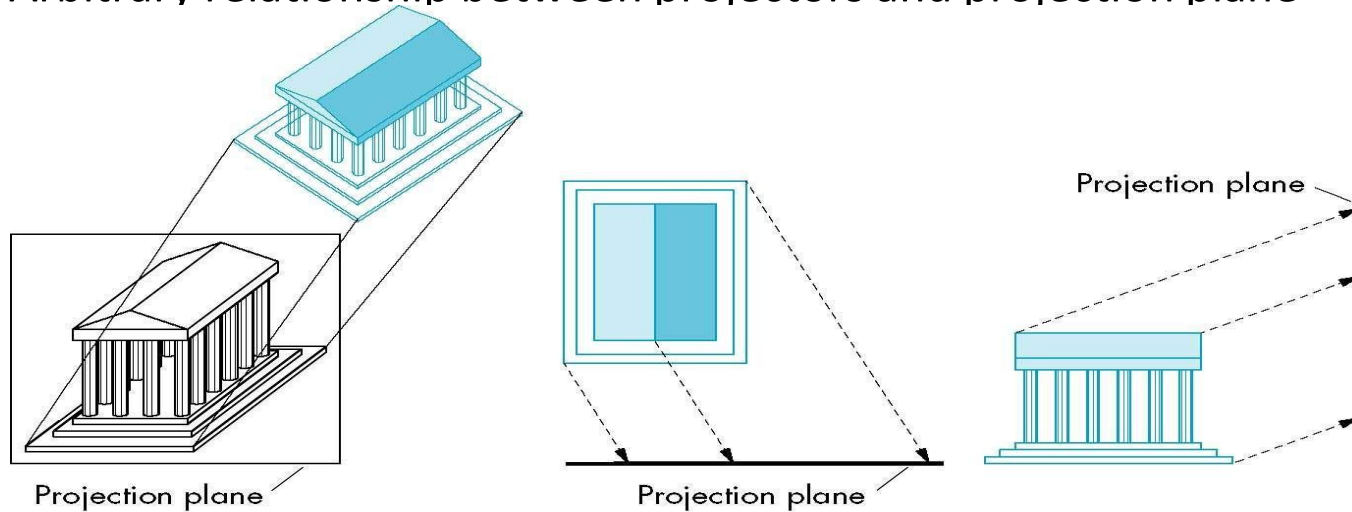


Advantages and Disadvantages

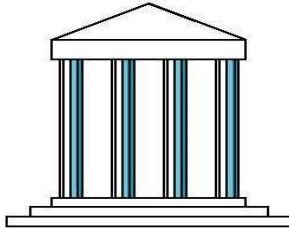
- Lines are scaled (*foreshortened*) but can find scaling factors
- Lines preserved but angles are not
 - Projection of a circle in a plane not parallel to the projection plane is an ellipse
- Can see three principal faces of a box-like object
- Some optical illusions possible
 - Parallel lines appear to diverge
- Does not look real because far objects are scaled the same as near objects
- Used in CAD applications

Oblique Projection

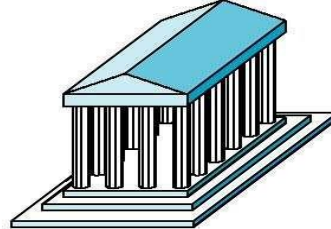
Arbitrary relationship between projectors and projection plane



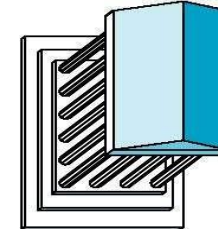
Classical Projections review



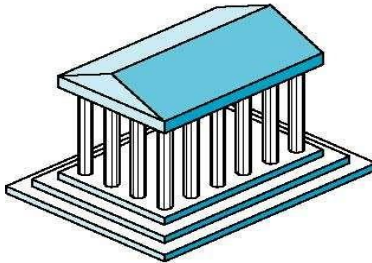
Front elevation



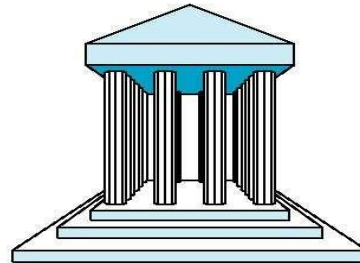
Elevation oblique



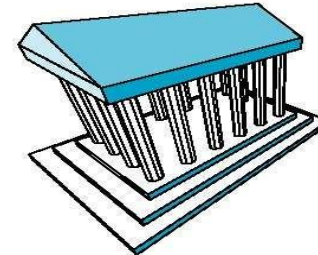
Plan oblique



Isometric



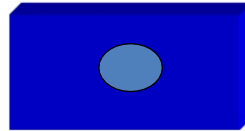
One-point perspective



Three-point perspective

Advantages and Disadvantages

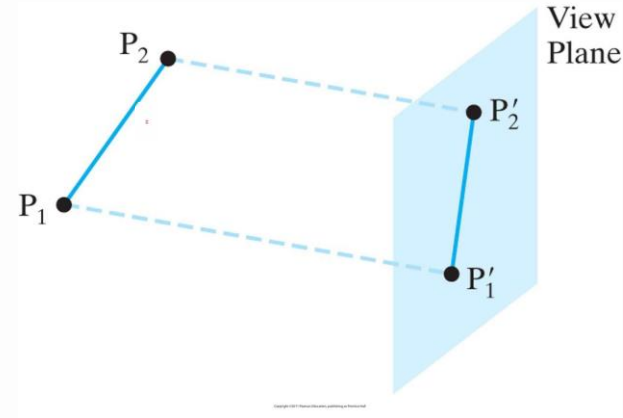
- Can pick the angles to emphasize a particular face
 - Architecture: plan oblique, elevation oblique
- Angles in faces parallel to projection plane are preserved while we can still see “around” side



- In physical world, cannot create with simple camera; possible with bellows camera or special lens (architectural)

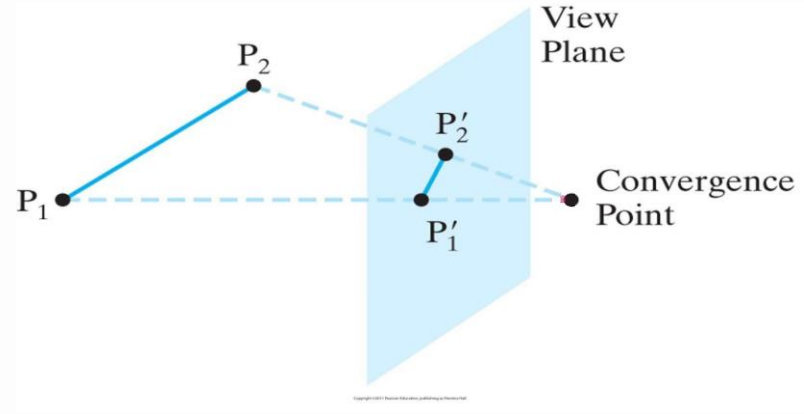
Projection Transformation (Contd..)

- Basic types of projections
 - **parallel projection**, where proportions are preserved
- There are **two general methods** for obtaining a parallel-projection view of an object:
 - We can project along lines that are perpendicular to the view plane,
 - or we can project at an oblique angle to the view plane.



Projection Transformation (Contd..)

- Basic types of projections
 - **perspective projection**, which matches how you see things in daily life. Perspective makes objects that are farther away appear smaller; for example, it makes railroad tracks appear to converge in the distance





Projection Transformation (Contd..)

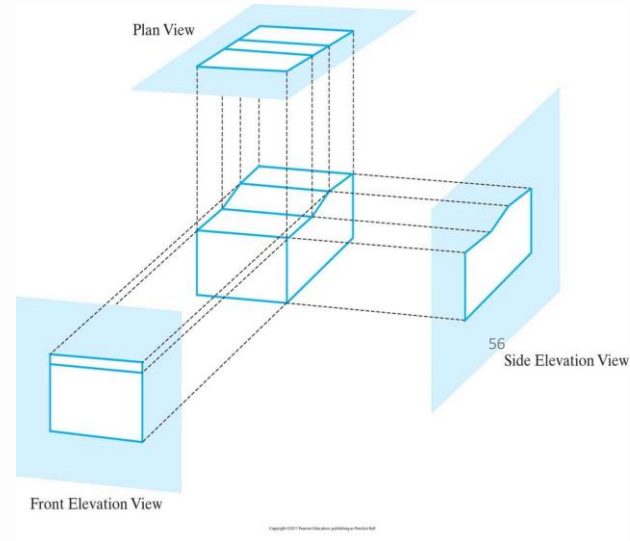
- **Basic types of projections**
 - **orthographic**, which maps objects directly onto the screen without affecting their relative size. Orthographic projection is used in architectural and computer-aided design applications where the final image needs to reflect the measurements of objects rather than how they might look
 - **Elevations:** Side, front, rear orthogonal projection
 - **Plan view:** Top orthogonal projection



Projection Transformation (Contd..)

Example:

*Orthogonal projections of an object,
displaying plan and elevation views.*



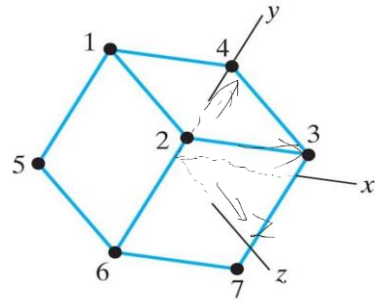


Orthographic Projections

- Axonometric and Isometric orthogonal projections
 - **Axonometric Orthogonal projections** display more than one face of an object
 - The most used axonometric projection type is **isometric projection**

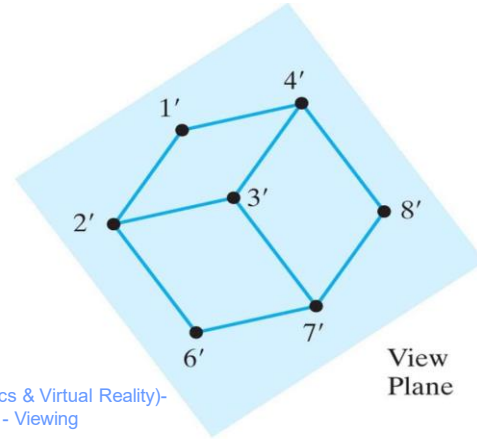
Isometric Projections

- Isometric projection is obtained by aligning the projection or the object, so that the plane intersects each coordinate axis (called principal axes) in which the object is defined.



Computer Graphics & Virtual Reality)-
Unit IV - Viewing

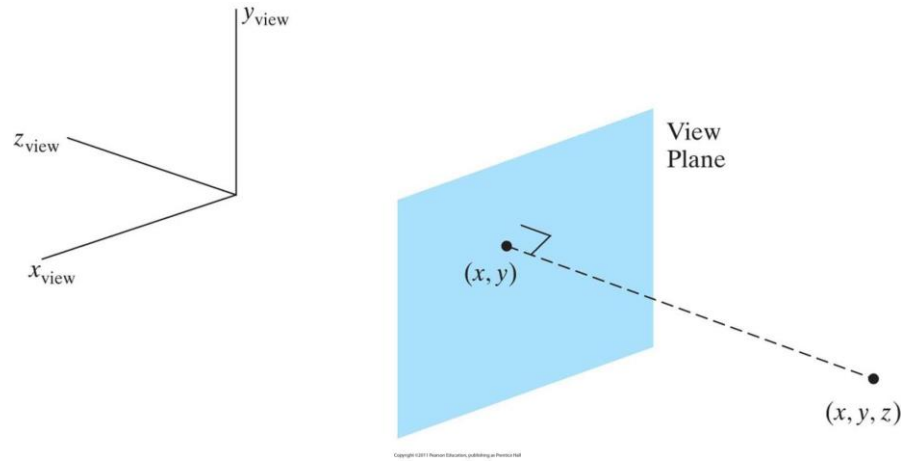
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Orthogonal Projection Coordinates

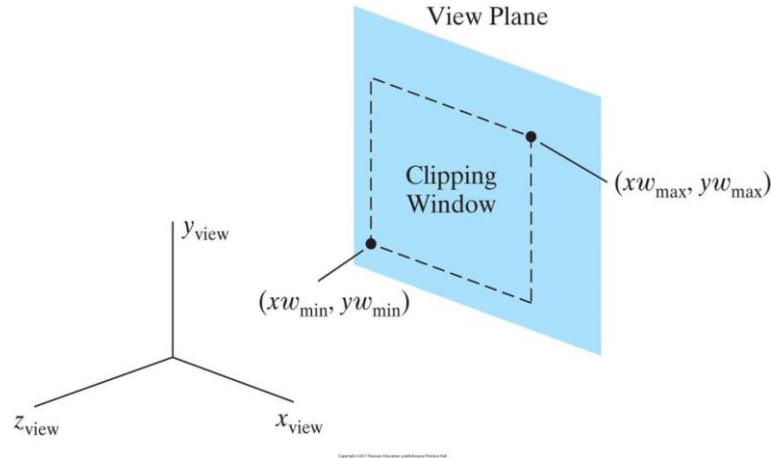
- With projection direction parallel to z_{view} axis, all transformation equations are trivial, i.e.

$$x_p = x \quad \text{and} \quad y_p = y$$



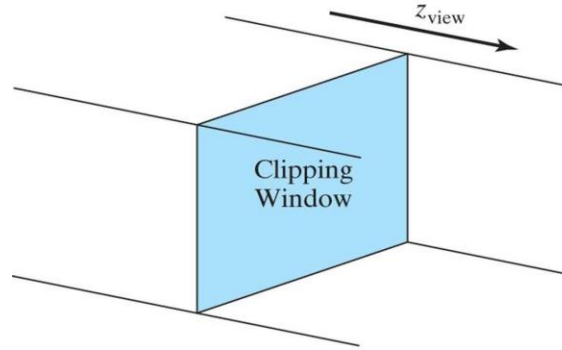
Clipping Window and Orthogonal Projection

- In CG generally rectangular clipping windows are used.
- In OpenGL allow clipping rectangles in standard position

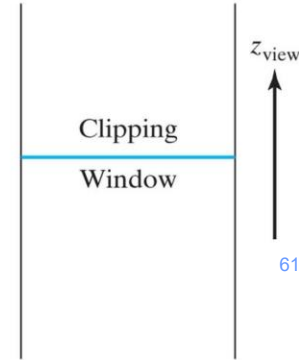


Clipping Window and Orthogonal Projection (cont.)

- Limiting the volume with near-far or front-back clipping planes



Side View
(a)

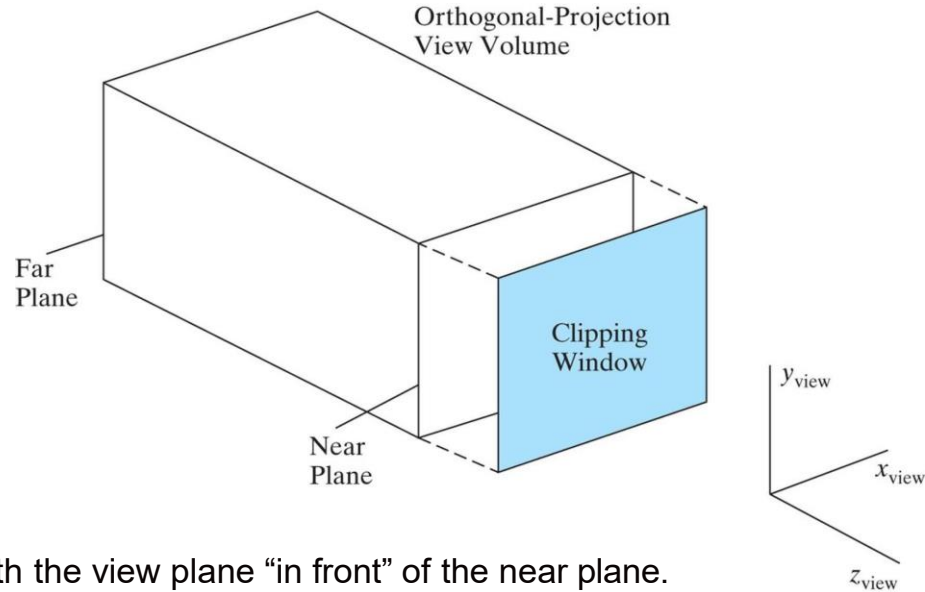


Top View
(b)

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Finite Orthogonal View Volume



with the view plane “in front” of the near plane.

Other Projections

- Oblique Parallel Projections
(other type)
- Perspective Projections

Viewport Transformation and 3D Screen Coordinates

- Similar to 2D

$$M_{normviewvol,3Dscreen} = \begin{bmatrix} \frac{xv_{max} - xv_{min}}{2} & 0 & 0 & \frac{xv_{max} + xv_{min}}{2} \\ 0 & \frac{yv_{max} - yv_{min}}{2} & 0 & \frac{yv_{max} + yv_{min}}{2} \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

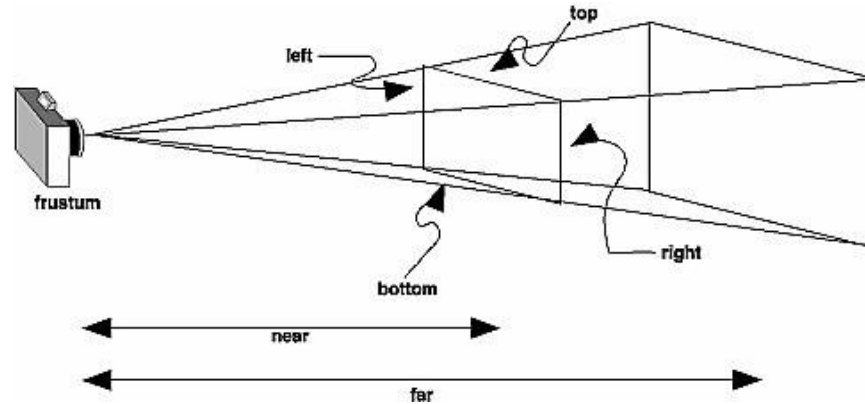
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Projection Transformation

- Two basic types of projections are provided by OpenGL
 - **Perspective projection**
 - **Orthographic / orthogonal projection**

Perspective Projection

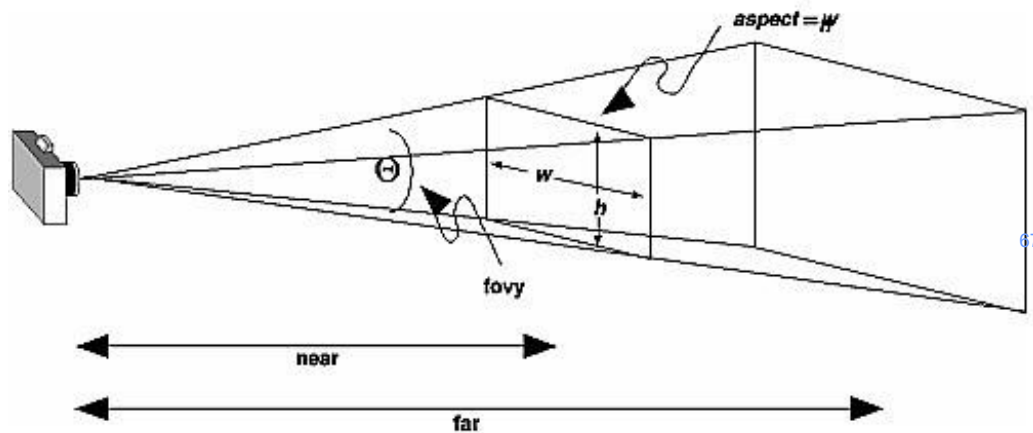
- `glFrustum(GLdouble left, GLdouble right, GLdouble bottom, GLdouble top, GLdouble near, GLdouble far)`



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Perspective Projection (cont.)

- `gluPerspective(GLdouble fovy, GLdouble aspect, GLdouble near, GLdouble far)`

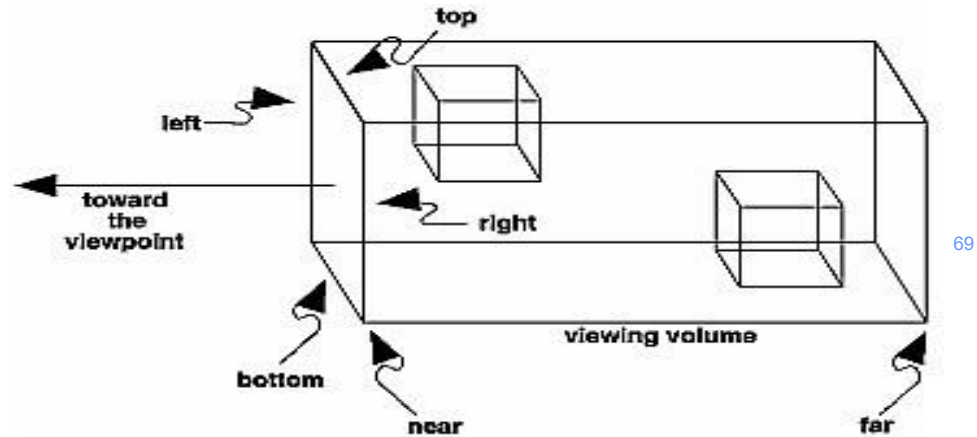


Projection Transformation

- `glMatrixMode (GL_PROJECTION);`
`glLoadIdentity ();`
`glFrustum (-1.0, 1.0, -1.0, 1.0, 1.5, 20.0);`

Orthographic Projection

- `glOrtho(GLdouble left, GLdouble right, GLdouble bottom, GLdouble top, GLdouble near, GLdouble far)`



Viewport Transformation

- Together, the projection transformation and the viewport transformation determine how a scene gets mapped onto the computer screen
- The projection transformation specifies the mechanics of how the mapping should occur, and the viewport indicates the shape of the available screen area into which the scene is mapped

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Viewport Transformation (cont.)

- `glViewport(GLint x, GLint y, GLsizei width, GLsizei height);`
 - The (x, y) parameter specifies the lower-left corner of the viewport, and width and height are the size of the viewport rectangle

Drawing the Scene

- Once all the necessary transformations have been specified, you can draw the scene (that is, take the photograph)
- As the scene is drawn, OpenGL transforms each vertex of every object in the scene by the modeling and viewing transformations.
- Each vertex is then transformed as specified by the projection transformation and clipped if it lies outside the viewing volume described by the projection transformation.
- Finally, the remaining transformed vertices are divided by w and mapped onto the viewport.

3D Clipping

- Main idea: Extending 2D thoughts into 3D
- Assumption: The boundaries are normalized
- View volume can be normalized to unit cube or to a symmetric cube with edge length 2 □ clipping planes at 0 and 1 or -1 and 1
- If clipping window normalized to symmetric cube:

$$xW_{min} = -1 \quad xW_{max} = 1$$

- Clipping plane equations: $yW_{min} = -1 \quad yW_{max} = 1$

$$zW_{min} = -1 \quad zW_{max} = 1$$

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3D Clipping (cont.)

- Remember: Homogeneous coordinate representation; $(x, y, z) \rightarrow (x, y, z, 1)$

- So:
$$\begin{bmatrix} x_h \\ y_h \\ z_h \\ h \end{bmatrix} = M \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$
 where M represents the concatenation of all various transformations from world to normalized, homogeneous projection coordinates.

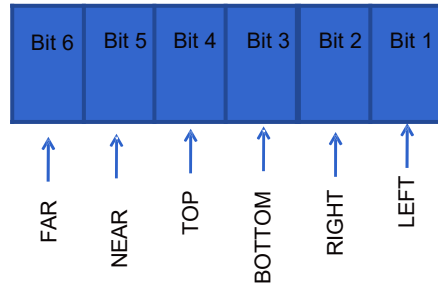
- Problem: h can be $\neq 1$

3D Clipping (cont.)

- If $h=1$ homogeneous parameters=Cartesian projection parameters, e.g. Parallel projection
- If $h \neq 1$ and clipping is done after division by h some coordinate information can be lost and objects may not be clipped correctly
- How to solve this problem?
- Answer: Apply clipping to homogeneous coordinate representation of spatial positions
- But how?
- See next slides

3D Clipping (cont.)

- 3D region codes: (Extending Cohen-Sutherland's idea to 3D)



3D Clipping (cont.)

- How to find out if a point is inside or outside the volume?
- Assumptions:
 - Each point in the 4-component representation: $P = (x_h, y_h, z_h, h)$
 - Clipping against normalized symmetric cube (See eq. Slide 44)
- Given these assumptions a point is inside the volume if the projection coordinates of the point satisfies the following equations:

$$-1 \leq \frac{x_h}{h} \leq 1, \quad -1 \leq \frac{y_h}{h} \leq 1, \quad -1 \leq \frac{z_h}{h} \leq 1$$

3D Clipping (cont.)

- Assuming $h \neq 0$

$$-h \leq x_h \leq h, \quad -h \leq y_h \leq h, \quad -h \leq z_h \leq h \quad \text{if } h > 0$$

$$h \leq x_h \leq -h, \quad h \leq y_h \leq -h, \quad h \leq z_h \leq -h \quad \text{if } h < 0$$

- So what?

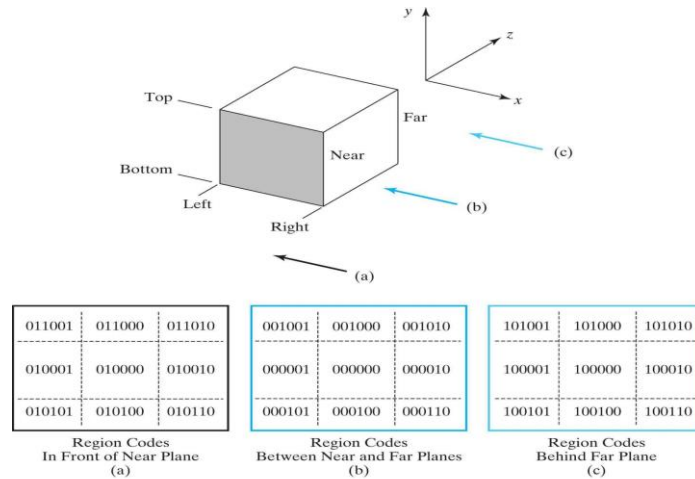
3D Clipping (cont.)

- Testing region codes (for $h > 0$):

<i>bit 1 = 1</i>	if $h + x_h < 0$	(left)
<i>bit 2 = 1</i>	if $h - x_h < 0$	(right)
<i>bit 3 = 1</i>	if $h + y_h < 0$	(bottom)
<i>bit 4 = 1</i>	if $h - y_h < 0$	(top)
<i>bit 5 = 1</i>	if $h + z_h < 0$	(near)
<i>bit 6 = 1</i>	if $h - z_h < 0$	(far)

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3D Clipping (cont.)



Values for the three-dimensional, six-bit region code that identifies spatial positions relative to the boundaries of a view volume.

3D Clipping (cont.)

- And for $h < 0$?
- Think about it

3D Clipping Algorithm

- For point clipping:
 - Eliminate all points with region code $\neq 000000$
- For line clipping:
 1. Test trivial accepts and trivial rejects using region codes, e.g. for trivial accepts use logical OR and for trivial rejects use logical AND
 2. For other cases, use methods of Cyrus-Beck or Liang- Barsky.
- 3D Polygon and curve clipping

3D Clipping in OpenGL

- In OpenGL optional clipping planes can be defined and used
- Defining and activating clip planes:
`glClipPlane (id, planeParameters );`
`glEnable (id);`
- Deactivating:
`glDisable (id);`
- Plane parameters: A, B, C, D
- Remember: Plane equation: $Ax+By+Cz+D=0$

3D Clipping in OpenGL

- A point (x, y, z) with $Ax+By+Cz+D<0$ will be eliminated
- Why?
- Because it is 'behind' the clipping plane

OpenGL 3D Viewing Functions

TABLE 10 - 1

Summary of OpenGL Three-Dimensional Viewing Functions

Function	Description
<code>gluLookAt</code>	Specifies three-dimensional viewing parameters.
<code>glOrtho</code>	Specifies parameters for a clipping window and the near and far clipping planes for an orthogonal projection.
<code>gluPerspective</code>	Specifies field-of-view angle and other parameters for a symmetric perspective projection.
<code>glFrustum</code>	Specifies parameters for a clipping window and near and far clipping planes for a perspective projection (symmetric or oblique). 85
<code>glClipPlane</code>	Specifies parameters for an optional clipping plane.

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Thank you